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Instruction Manual
ID-442 Infrared Detector
442-1A Temperature Controller
442-2 Preamplifier

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I. General Descriptions:

The Princeton Instruments ID-442 is an infrared detector assembly for the wavelength range of 1.1 to 2.9 microns. It comes complete with housing and mounting for the Acton monochromators and includes a fixed bias resistor. It also includes a single stage thermoelectric cooler capable of providing a stable operating temperature to –20 degrees Centigrade when used with a temperature controller such as the Princeton Instruments 442-1A. A thermistor is provided to monitor the detector temperature and for controlling the detector cooling.

The 442-1A is the optional temperature controller for the ID-442IR detector. It provides the controlled power for the thermoelectric cooler in the detector and allows setting the detector operating temperature by adjusting the thermistor resistance set point. Temperature stability is better than 0.1 degree Centigrade. Both heating and cooling is provided.

The 442-2 is the optional preamplifier for the ID-442 IR detector. The preamplifier provides a fixed gain of 100 for the detector output. It is contained in the same enclosure as the detector and signal leads are kept short to minimize noise pickup. When ordered with the 442-1 Temperature Controller, the Amplifier and Bias Power Supplies are contained in the 442-1. Otherwise, the power supplies are provided in a separate module.

II. Specifications

Model ID442: IR Detector

Type:	PbS
Active Area:	5 x 5 mm
Wavelength Range:	1.1 to 2.9 microns
D* (Peak, 750,1):	1×10^{11} (see curve in appendix)
Max. current for thermoelectric cooler:	1.8 Amps @ 0.8 Volts

Model 442-1A: Temperature Controller

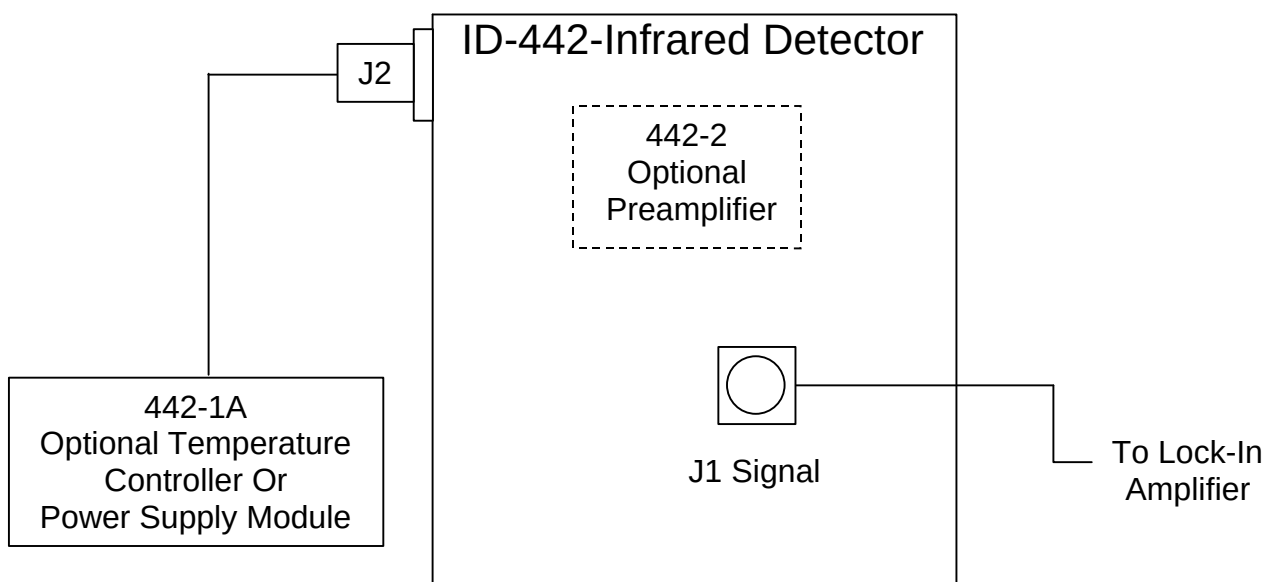
Temperature Stability:	Better than 0.1 degree Centigrade
Provides up to 2.5 Amps of current for the thermoelectric cooler (When used with ID-442 Detector, set Current Limit to 1.8 Amps)	

Model 442-2: Preamplifier

Gain:	100 (fixed)
Bandwidth:	70 Hz to 20 KHz
Noise:	870 nV/Hz
Output voltage (max):	20 V pk – pk

III. Operation

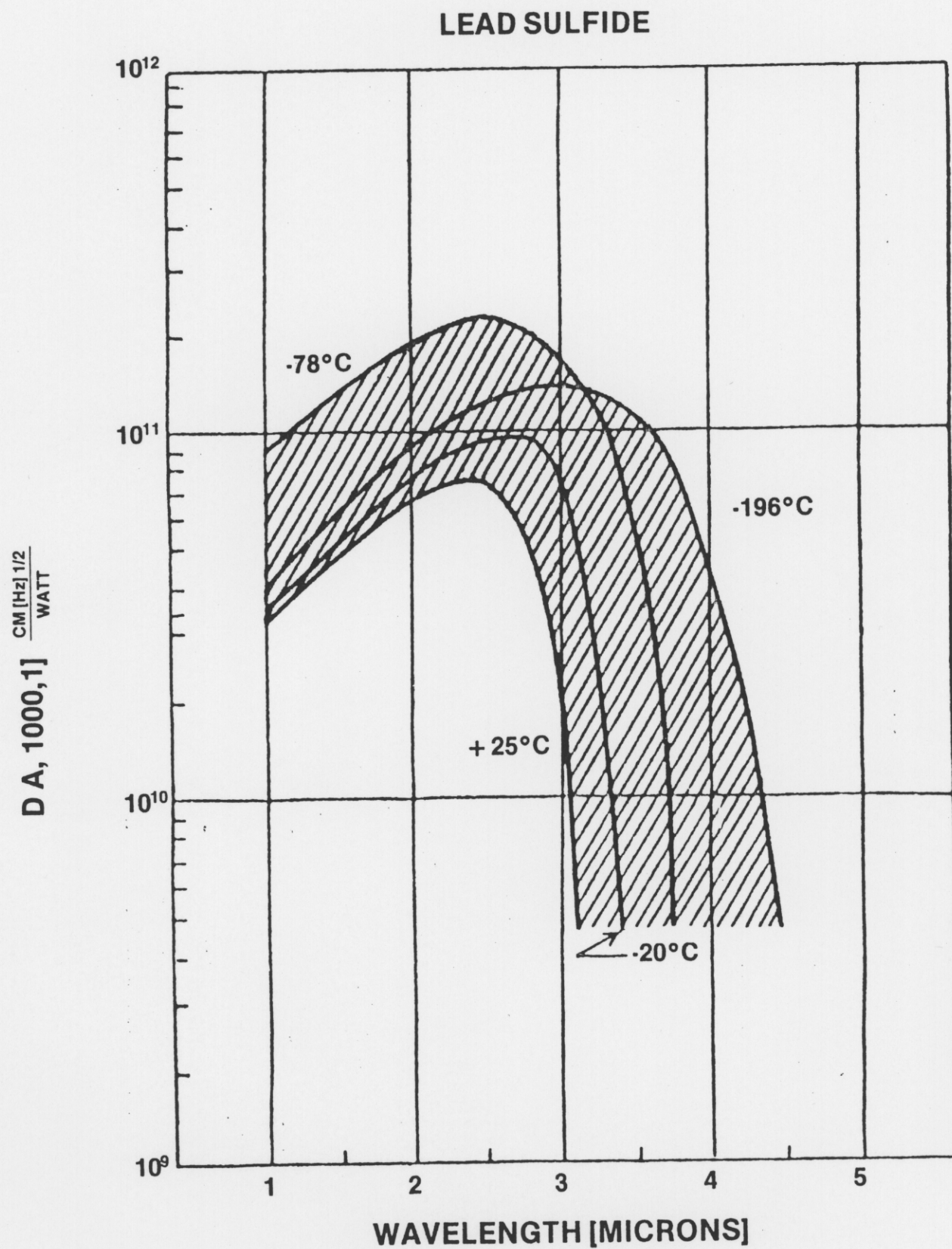
Mount the ID-442 IR detector assembly to the exit slit assembly of the Acton monochromator with the four cap screws provided and with the connectors on the ID-442 toward the front of the monochromator. Connect the ID –442 IR detector assembly and optional 442-1A temperature controller as shown in Figure 1 below. The optional ID-442-2 preamplifier is factory installed inside the ID-442 detector assembly when purchased.



The detector bias and preamplifier draw less than 10 mA of power supply and current and the power supply may remain connected to the detector with no effect on the life of the detector or preamplifier.

The detector and preamplifier are meant to be operated with a chopped or alternating signal. The preamplifier output is typically connected to a lock-in amplifier. If using the 442-1A temperature controller, set the "TEC CURRENT MAX SSET" to 1.8 Amps, the "TEC TEMPERATURE SET" to 7.52. This value of thermistor resistance corresponds to -20°C for this particular unit. If using a temperature controller other than the Princeton Instruments 442-1A, be certain not to exceed the current and voltage ratings of the thermoelectric cooler (1.8 Amps @ 0.8 Volts).

IV. Appendix A: PbS Characteristic Curve



IV. Appendix B: Typical Resistance Values for 1K Thermistor

Temp (°C)	Thermistor K	Temp (°C)	Thermistor K
0	2.558	0	2.559
-1	2.666	1	2.456
-2	2.779	2	2.358
-3	2.897	3	2.264
-4	3.021	4	2.175
-5	3.151	5	2.039
-6	3.288	6	2.008
-7	3.431	7	1.930
-8	3.581	8	1.856
-9	3.739	9	1.784
-10	3.905	10	1.716
-11	4.079	11	1.651
-12	4.261	12	1.589
-13	4.453	13	1.530
-14	1.654	14	1.473
-15	4.866	15	1.419
-16	5.088	16	1.366
-17	5.322	17	1.317
-18	5.567	18	1.269
-19	5.826	19	1.223
-20	6.097	20	1.180
-21	6.383	21	1.138
-22	6.683	22	1.098
-23	6.999	23	1.059
-24	7.332	24	1.022
-25	7.681	25	0.987
-26	8.049	26	0.953
-27	8.437	27	0.920
-28	8.845	28	0.889
-29	9.275	29	0.859
-30	9.727	30	0.831
-31	10.204	31	0.803
-32	10.705	32	0.776
-33	11.235	33	0.751
-34	11.792	34	0.726
-35	12.379	35	0.703
-36	12.998	36	0.680
-37	13.651	37	0.658
-38	14.339	38	0.637
-39	15.065	39	0.617
-40	15.830	40	0.598

IV. Appendix C: Connector Pin Descriptions:

J1: Signal (BNC Connector)

J2: Power / Thermoelectric Cooler

1	thermoelectric cooler +
2	thermoelectric cooler –
3	thermistor
4	thermistor
5	open
6	open
7	+15 Volts DC 25mA
8	–15 Volts DC 25mA
9	ground